

A073 · PerturbRx: Treatment-Conditioned Latent Transitions for Patient Drug Response Prediction

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Session: Day 2 · Fri Oct 2, 2026 · 2:15–4:15 PM

Patient-level cancer treatment-response prediction remains challenging because clinical response data are limited and matched pre- and post-treatment molecular profiles are rarely available. We introduce PerturbRx, a treatment-conditioned representation learning framework that leverages large-scale single-cell perturbation data to learn latent transitions for patient drug-response prediction. PerturbRx first learns a drug- and dose-conditioned transition predictor in a shared pretrained latent space using context-matched but unpaired control and perturbed populations from the Tahoe-100M single-cell perturbation atlas. The pretrained predictor is then frozen and applied to pretreatment patient transcriptomic profiles to generate patient- and drug-conditioned latent transitions without requiring matched post-treatment measurements. On Tahoe-100M, PerturbRx outperformed identity, global-transition, linear, MLP, and conditional autoencoder baselines in predicting treatment-induced latent transitions, achieving a maximum mean discrepancy (MMD) of 0.0682 and a transition cosine similarity of 0.6715. For patient response prediction on a TCGA benchmark comprising 508 treatment episodes from 462 patients, PerturbRx achieved the highest overall performance among the four evaluated drug-response methods, with an AUROC of 0.692 ± 0.055 and an AUPRC of 0.787 ± 0.055 . These results suggest that perturbation-pretrained latent transitions provide useful representations for patient-level drug-response prediction and offer a practical strategy for transferring single-cell perturbation information to settings where only pretreatment patient profiles are available.

